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Artificial neural networks for prediction of recurrent venous thromboembolism

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ABSTRACT

Background: Recurrent venous thromboembolism (RVTE) is a multifactorial disease with occurrence rates which vary from 13 % to 25 % in 5 years after the initial event. Once a patient the first thrombotic event, the probability of recurrence should be determined, as well as the most adequate anticoagulant therapy. To our knowledge based on the published literature, three statistical models have been proposed to calculate RVTE probability. However, these models present several limitations, such as: limited input variables, lack of external validation and applicability only for patients with a first unprovoked thrombosis. Additionally, some of the models have been recognized to fail in determining RVTE when patients have a low risk of recurrence.
Objective: An alternative procedure in which three Artificial Neural Network (ANN) models were developed to classify which patients will present RVTE based solely on clinical data.
Methods: Data of 39 clinical factors from 235 patients were used to train several ANN structures. The difference among the three models was its inputs. In ANN 1, the inputs were all 39 factors. In ANN 2, 18 factors determined previously as the main predictors of RTVE using Principal Component Analysis (PCA). Finally, in ANN 3, 15 factors combining PCA results with practical aspects. Different number of hidden layers and neurons, and three optimization algorithms were considered. 5-fold cross validation was also performed.
Results: The results showed that all models were capable of performing this task. Different optimization algorithms lead to different results. The best models presented high accuracy. The best structures were 39 − 10-10 − 1, 18 − 10-5 − 1, and 15 − 15-10 − 1 for ANN 1, ANN 2, and ANN 3 models, respectively. The cross-validation showed that the results are consistent.
Conclusions: This work showed that the association of multivariate techniques and ANNs is a powerful tool that can be used to model a complex phenomenon such as RVTE without the restrictions of existing approaches.
Application: After proper validation, these ANN models can be used to help clinicians with decisions regarding VTE treatment.

1. Introduction

Venous thromboembolism (VTE) is a multiple factor disease and anticoagulation therapy is the default treatment [1]. However, once initiated, the question to be answered is how long patients should be treated. General guidelines usually recommend at least three months of treatment or indefinite anticoagulation therapy for patients presenting low-risk of bleeding and moderate to high risk of recurrence [1]. However, this subject remains controversial.

Scores have been proposed to calculate the risk of a patient develop recurrent venous thromboembolism (RVTE) [2–5]. Each score takes different prediction factors into consideration and can be used only in patients who had a previous unprovoked VTE. Recent reviews showed that these score models have a significant limitation when predicting patients with low risk of RVTE and still lack complete validation [6, 7].

As far as mathematical tools are concerned, Artificial Neural Networks (ANNs) represent a powerful tool that can relate dependent to independent variables with no prior knowledge regarding the relation among them and capable of dealing with noise data [8, 9]. This can be attained solely by knowing which independent factors are important and presenting this information to the ANN.

ANNs consist in a set of artificial neurons that are interconnected

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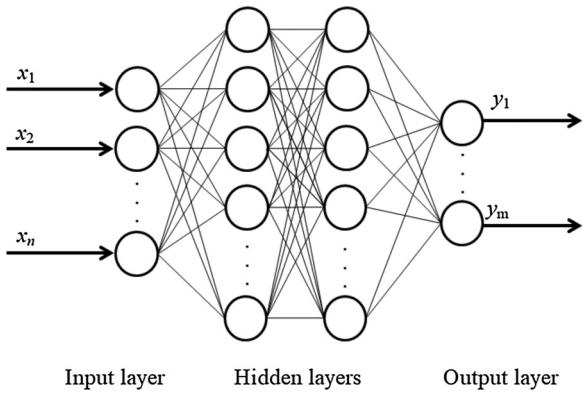

Fig. 1. General two hidden layer ANN structure. x_i is the input variable, y_i is the output variable, n and m are the total number of inputs and outputs, respectively.

and organized into layers. An ANN consists of input layer, hidden layers (one or more depending upon the requirements) and the output layer. The information process in an artificial neuron begins relating the input (I_i) to a synaptic weight (w_{ij}) that assesses its influence on the output of this neuron. Additionally, a bias parameter (b_j) is assigned to each ANN neuron with the purpose of obtaining a more versatile model. The sum of all these parameters originates the neuron activation (Eq. (1)) and the answer of the input stimulus is obtained applying an activation function to the neuron activation, generally the sigmoid or the hyperbolic tangent function. This procedure extends to the exiting layer of the network, where the neuron answer is the dependent variable of the problem (y_i) [9].

$$\alpha_j = \sum_{i=1}^{Z} w_{ij} I_i + b_j \tag{1}$$

where α_j is the neuron activation and Z is the total number of inputs for the neuron j. Fig. 1 shows the general architecture of an ANN with two hidden layers.

Recently, ANNs have been used with success as clinical decision support systems to predict several diseases, including: dengue, chronic myeloid leukemia, acute cerebral ischemia, portosplenomesenteric venous thrombosis, portal vein thrombosis, heart diseases, asthma, etc [10–19]. A recent review from Parveen et al. [20] shows several applications and advances concerning the application of ANNs in the medical field.

Fei et al. [11], trained radial basis ANNs to predict the risk for portal vein thrombosis in acute pancreatitis patients and obtained accurate results. Fei et al. [12] used a neural model to predict the incidence of portosplenomesenteric venous thrombosis in patients with acute pancreatitis. The authors reported that an ANN with 9 hidden layers was more accurate than the logistic regression model. In another study, Qatawneh et al. [21] trained several ANN structures to predict the risk of VTE in hospitalized patients. The authors used 35 input variables, such as: patients and disease characteristics, as well as genetic factors. The results showed that the ANNs have 81 % of accuracy. Recently, Willan et al. [22] also reported that ANNs can be used to classify patients with predisposition to VTE. A recent review showed that there are several studies that use ANNs for VTE prediction [23]. However, the RVTE prediction remains an unexplored field.

When working with ANNs, data reduction is a common practice, especially when a complex phenomenon with several variables is the problem to be treated. Principal Component Analysis (PCA) is a multivariate statistical method that identifies patterns and classifies the factors that influences a given phenomenon. PCA can also be used to reduce dimensionality, by cutting the variables that are less important. The technique is widely used to identify patterns in the medical field [24–28], and was also used to identify the main predictors of RVTE [29].

Bearing all this in mind, the aim of this work is to obtain an ANN to predict RVTE in patients with previous provoked and unprovoked VTE using clinical factors as inputs; in addition to associating PCA with ANN modeling in an attempt to obtain a model with less inputs and similar accuracy. The main contribution of this work is to show that the association of PCA with artificial intelligence is a powerful strategy to treat complex phenomenon such as RTVE.

2. Methods

When working with ANNs, one must find: i) the optimal number of hidden layers, ii) the optimal number of neurons in each hidden layer, and iii) the suitable weights for each ANN neuron of the phenomenon. This process is called *training*. A second step called *validation* is performed along with the training to avoid overfitting. Once the ANN is trained, a last step consists in testing the model using new data. This step is called *test*. If the errors in validation and test steps are low enough, the ANN is ready for use.

As part of the new procedure, three ANNs were proposed in this work. The first one is a model that includes all factors presented in Table 1 as inputs (ANN 1), which includes the clinical, inherited and acquired molecular factors, as well as molecular risk factors. The second ANN is a simplified model that uses as inputs some important factors determined by PCA in a previous work [29] (ANN 2). The inputs of the third ANN combine PCA factors with daily-based information (ANN 3). In the following sections, details about the development of each model are presented.

Table 1
RVTE factors considered in this work.

Selected Factors		
Sex (men/women)	G20210A prothrombin gene mutation (No/Yes)	Red blood cell (RBC) count (μL^{-1})
Age (years)	Factor VIII ($IU.dL^{-1}$)	Hemoglobin (HB) ($g.dL^{-1}$)
Pulmonary embolism (PE) (No/Yes)	Protein C level (PC) ($mg.dL^{-1}$)	Hematocrit (HCT) (%)
Side of inferior member (left/right)	Protein S level (PS) ($mg.dL^{-1}$)	Red Blood Cell Distribution (RDW) (%)
Level of VTE in leg (distal/ proximal)	Antithrombin level (AT) ($mg.dL^{-1}$)	Platelet (PLT) (μL^{-1})
Provoked or unprovoked	Antiphospholipid syndrome (No/Yes)	Mean Platelet Volume (MVP) (fL)
Anticoagulant treatment duration (months)	Body mass index (BMI)	Total cholesterol ($mg.dL^{-1}$)
Anticoagulant use (No/Yes)	Tobacco abuse (No/Yes)	High-density Lipoprotein (HDL) ($mg.dL^{-1}$)
D-dimer ($ng.\ mL^{-1}$)	Hormonal therapy (for women) (No/Yes)	Low-density Lipoprotein (LDL) ($mg.dL^{-1}$)
FV Leiden (No/Yes)	White blood cell (WBC) count (μL^{-1})	Triglycerides ($mg.dL^{-1}$)
Fasting blood glucose ($mg.dL^{-1}$)	Creatinine ($mg.dL^{-1}$)	C-reactive protein ($mg.dL^{-1}$)
Arterial hypertension (No/Yes)	Diabetes mellitus (No/Yes)	Dyslipidemia (No/Yes)
Renal failure (No/Yes)	Cancer (No/Yes)	Residual venous thrombus on US Doppler (No/Yes)

2.1. Population

This was a retrospective study and all collected data is published elsewhere [29]. In summary, 261 patients with a first provoked or unprovoked VTE seen between January 2009 and August 2016, assisted at outpatient clinic of Hemocentro Unicamp were selected. From the total number of patients, 26 were excluded as they presented missing data, thrombosis that occurred in other sites than pulmonary, lower limbs or central nervous system, and age under 18 years. The 235-patient study included 74 men (31.48 %) and 161 women (68.51 %). Ninety-two patients (39 %) presented a first provoked VTE: 13 (5.53 %) pregnancy and postpartum, 62 (26.38 %) hormone therapy for contraception or hormonal replacement, 2 (0.85 %) surgery, 2 (0.85 %) air travel longer than 6 h and 13 (5.53 %) cancer. Factor V Leiden and G20210A prothrombin mutation was detected in 46 (19.57 %) patients.

The average time of anticoagulation after the first thrombotic episode was 10.35 months (and the maximum was 72). All patients were treated for a minimum of six months, regardless of provoked or unprovoked VTE. Seven patients with diagnosis of Antiphospholipid Syndrome were maintained under continuous anticoagulation after the first VTE episode.

The detection of a novel thrombotic episode at a different site from the primary event, or a new episode occurred at the same localization of previous thrombosis, was defined as RVTE. From all 235 patients, 49 (20.85 %) presented RVTE. The average time between thrombotic episodes, and the time between the end of anticoagulation therapy and the second thrombotic episode were: 23.75 ± 19.12 months and 15.87 ± 14.89 months, respectively. Among the 49 patients that presented RVTE, 27 of them presented a first provoked VTE, 22 presented a first unprovoked VTE, and 13 experienced RVTE during anticoagulation therapy (6 under continuous indication after first episode, and 7 during the period of anticoagulation higher than 6 months of therapy).

2.1.1. ANN 1

For ANN 1, all gathered data was used for training, validation and test. The 39 factors presented in Table 1 were used as inputs. The permitted range for each factor can be found in Table S1 at the Supplemental File. Since there are some input variables that are categorical data, they were converted into a numerical form, as shown in Table 2. All numerical variables were used without any unit conversions. The desired output was the occurrence of RVTE, which is also a categorical variable and was valued as 1 if positive, and -1 if negative.

Three different optimization methods were used to train the ANNs: Resilient Backpropagation [30], Levenberg-Marquardt [31], and Powell-Beale [32, 33]. Following Qatawneh et al. [21], the activation function in all layers was hyperbolic tangent. ANNs with one, two and three hidden layers were tested, and the number of neurons varied between 10 and 30 in the first hidden layer, and was 5 or 10 in the second hidden layer. The objective function minimized by the methods was the mean square error given by the Eq. (2):

$$F_{OBJ} = \sum_{i=1}^{m} \frac{(y - y')^2}{m} \tag{2}$$

where y is the real data value, y' is the ANN predicted data, and m is the total number of real data provided to the ANN. From all available data, 70 %, 15 %, 15 %, were used for training, validation and test steps, respectively. All input and output data were normalized in the range of [-1;1]. All simulations were performed using a script written in Mathworks MatLab®. After the simulations were performed, test and validation predictions was compared with the real ones to obtain the accuracy (ACC - Eq. (3)) of each ANN.

$$ACC = 100\frac{TR + TN}{TP} \tag{3}$$

where: TR is the number of correct RVTE predictions, TN is the number of correct non-RVTE prediction, and TP is the total number of patients (in this case: 235). In Figure S1, at the Supplemental File, is presented a flow diagram of the ANN 1 training.

Finally, since the ANN outputs are continuous variables, receiver operating characteristic (ROC) curves were constructed, and the area under the curve (AUC) was calculated for each trained structure.

2.1.2. ANN 2

In a recent study [29], we used PCA [34,35] to evaluate the main factors to predict RVTE. The results indicated that for our population 18 of all 39 factors are important for RVTE prediction, as depicted in Table 3. All variables are numerical, except the proximal or distal lower limb thrombosis, which was converted according to Table 2. Since these important factors are simple to obtain, in this work an association between the PCA and the ANN was proposed to obtain a model with 18 important factors used as inputs.

The Powell-Beale method was used to train the ANNs. All other setup configuration was the same as for ANN 1 (data normalization, activation functions, F_{OBJ} equation, number of hidden layers and neurons, number of patients, data division, and accuracy) and is given in the Population Section. Figure S2, at Supplemental File, shows a flow diagram of the ANN 2 training procedure.

2.1.3. ANN 3

Finally, in an attempt to obtain a simpler and more useful ANN model, we trained several ANN structures with a slightly change in the inputs of ANN 2. In this case, we removed the factors: anticoagulant treatment duration, protein C, protein S, and antithrombin from the inputs, and added the factor "provoked or unprovoked". It consisted in a 15 inputs ANN. This was done because not all hospitals determine PC, PS, and AT levels, to differentiate those patients with a previous provoked VTE from those with an unprovoked episode, and because the use of anticoagulant treatment duration as input is somewhat controversial. All the configuration setup was the same as for ANN 2.

Table 2
ANN inputs for non-numerical variables.

Factor	ANN input value
Sex	Male = 1; Female = -1
Pulmonary embolism	Yes = 1; No = -1
Side of inferior member	Right inferior member = 1 Left inferior member = -1
Level of VTE in leg	Proximal = 1; Distal = -1
Provoked or unprovoked	Provoked = 1; Unprovoked = -1
Anticoagulant use	Yes = 1; No = -1
FV Leiden	Negative = 1; Heterozygous = -1
G20210A prothrombin gene mutation	Negative = 1; Heterozygous = -1
Antiphospholipid syndrome	Negative = 1; Positive = -1
Tobacco abuse, hormonal therapy, arterial hypertension, diabetes mellitus, dyslipidemia, cancer, residual venous thrombus on US doppler	Yes = 1; No = -1

Table 3
Main RVTE factors used as inputs in ANN 2 model.

Selected Factors		
Age	Antithrombin level	Cholesterol
Level of VTE in leg	WBC	HDL
Anticoagulant treatment duration	RBC	LDL
D-dimer	HB	Triglycerides
Protein C level	HCT	Glycemia
Protein S level	RDW	Creatinine

2.1.4. Neurons saturation

During the training of each ANN structure, the degree of saturation was monitored to prevent overfitting and the vanishing gradient problem. Saturation occurs when the neurons in the hidden layers generate exit values close to the asymptotic endings of the activation function range [36,37]. The hyperbolic tangent presents an output range of [-1; +1]. If α_j, is too high, it will generate an output equal to 1. The output will be -1 when the α_j is too low. Thus, when a neuron is saturated, its output will be only +1 or -1 and any variation on the weighs does not influence the output value, compromising the ANN learning.

Rakitianskaia and Engelbrecht [36] proposed a method to determine the neurons saturation which is very simple to use. After the ANN learning, the outputs of the hidden layers neurons can be calculated and the frequency distribution of these values can be ploted in a histogram. Then, a single-value saturation measure, ψ_B, can be obtained using Eq. (4):

$$\Psi_B = \frac{\sum_{b=1}^{B} |\overline{g}_b'| \, f_b}{\sum_{b=1}^{B} f_b} \tag{4}$$

where:

$$\overline{g}_b' = \frac{2(\overline{g}_b - g_L)}{g_U - g_L} - 1 \tag{5}$$

$$\overline{g}_b = \begin{cases} \sum_{k=1}^{f_b} g_k / f_b & \text{if} f_b > 0 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

In Eqs. (4)–(6), B is the total number of bins, b is the bin index, f is the number of outputs in bin b, g is the hidden neuron output, U and L are the upper and lower bounds of g.

The saturation measure, ψ_B, can assume values in the range of [0;1]. If g presents an uniform distribution ψ_B will be 0.5 and the model does not present saturation. If the frequency of g is higher at the boundaries, φ_B will be close to 1.0, and this indicates that the model is saturated.

2.1.5. 5-Fold cross-validation

To ensure the ANNs prediction, we performed a 5-fold cross-validation scheme using the best ANN structure obtained for each proposed model. In this case, from all available data, 80 % was used to train the ANNs, and 20 % to test stage. In each fold run, the index range of the validation data was a different subset, so all data was in the validation data set once, as shown at Table 4. To evaluate the result of each k-fold, it was computed the accuracy for the validation dataset. ROC curves were constructed and the AUC was calculated for each k-fold run. Finally, the average accuracy was calculated as the mean of all individual k-fold accuracies.

Table 4
Cross-validation data selection for ANN 1 and ANN 2.

k-Fold Run	Training Index	Test Index
1	1 – 188	189 – 235
2	1 – 141 and 188 – 235	141 – 188
3	1 – 95 and 142 – 235	95 – 141
4	1 – 47 and 95 – 235	48 – 94
5	48 – 235	1 – 47

2.1.6. Inputs importance

To evaluate which input has more impact in each model, we calculated the inputs relative importance index using the cumulative weight score (CWS), proposed by O'Donoghue et al. [38]. In this method the cumulative weight for each input can be calculated using by the equation:

$$\boldsymbol{\varphi} = \prod_{j=1}^{L} \mathbf{w_j} \tag{7}$$

In Eq. (7), $\boldsymbol{\varphi}$ has dimension n x m, where n is the number of inputs and m is the number of outputs, $\mathbf{w_j}$ is the weight matrix of the layer j, and L is the total number of layers. The importance of each input is obtained by sorting the elements of each column of $\boldsymbol{\varphi}$ in descending order of absolute values, and then dividing each element of $\boldsymbol{\varphi}$ by the sum of all elements in a column.

3. Results

3.1. ANN 1

The ANN 1 model consisted of networks with 39 input variables and one output, the positive/negative response for RVTE. Three optimization algorithms were used to train several ANN structures. Table 5 presents the F_{OBJ} for training, validation, and, test steps obtained using the three optimization algorithms considered. Also, the accuracy and AUC are presented, for validation and test steps, for each structure.

The results showed that the F_{OBJ} values for training step do not follow any general tendency as the number of neurons increases. Furthermore, the performance of each optimization algorithm is different, as well as the accuracy of each ANN structure. In general, when comparing training F_{OBJ} value, the ANNs trained with Levenberg-Marquardt algorithm presented the worst performance and the best F_{OBJ} values were obtained when using the Powell algorithm.

When analyzing the accuracy values, the tendency remains the same: the best results were obtained when using Resilient Backpropagation and Powell algorithms. The accuracy when using the Levenberg-Marquardt algorithm were much lower in comparison with the other ones.

The best results for each algorithm were obtained by different ANN architectures. The optimal result was obtained by ANN 39 – 10-10 – 1, using Powell-Beale algorithm, with two hidden layers. The F_{OBJ} value for training, validation, and test steps was 0.202, 0.021, and 0.158, respectively. The accuracy obtained was 92.8 %. This result shows the importance of using different optimization algorithms and testing several ANN structures to find the optimal solution.

3.2. ANN 2 and ANN 3

The ANN 2 model consisted in networks with 18 input variables. The ANN 3 model consisted in networks with 15 input variables. The output was the positive/negative response for RVTE. Due to its performance, only the Powell-Beale algorithm was used to train the ANN structures. Table 6 presents the F_{OBJ} for training, validation, and, test steps. Also, the accuracy and AUC are presented, for validation and test steps, for each structure.

The results presented in Table 6 show that the training F_{OBJ} values presented no general tendency either as the number of neurons increases for both models. Furthermore, these values were higher in comparison with the ANN 1 models trained with the same algorithm. The best model for ANN 2 was the structure with 10 and 5 neurons in the first and second hidden layers, respectively. For ANN 3, the structure with 15 and 10 neurons in each layer was the best. The F_{OBJ} value for training, validation, and test steps was similar to ANN 1. The

Table 5
Summary of the results obtained with ANN 1 model.

Structure	F_{OBJ}Training	F_{OBJ}Validation	F_{OBJ} Test	Validation + Test Accuracy (%)	Validation + Test AUC	$\psi_{B,1}$	$\psi_{B,2}$
Resilient Backpropagation							
39–10-1	0.297	0.285	0.352	65.7	0.893	0.952	-
39–30-1	0.222	0.289	0.412	65.7	0.738	0.961	-
39–35-1	0.364	0.204	0.522	81.4	0.892	0.856	-
39–5-5–1	0.159	0.174	0.853	71.6	0.863	0.896	0.870
39–10-5–1	0.297	0.437	0.412	35.1	0.715	0.864	0.832
39–15-5–1	0.359	0.464	0.460	52.2	0.708	0.927	0.893
39–20-5–1	0.263	0.152	0.751	71.6	0.797	0.905	0.820
39–30-5–1	0.185	0.152	0.210	73.8	0.860	0.870	0.815
39–5-10–1	0.371	0.309	0.418	79.2	0.901	0.893	0.870
39–10-10–1	0.251	0.446	0.480	84.2	0.952	0.892	0.822
39–15-10–1	0.126	0.011	0.056	84.6	0.963	0.915	0.897
39–20-10–1	**0.185**	**0.023**	**0.097**	**88.7**	**0.891**	**0.840**	**0.804**
39–30-10–1	0.196	0.233	0.259	86.4	0.899	0.895	0.831
Levenberg-Marquardt							
39–10-1	**0.224**	**0.166**	**0.427**	**84.3**	**0.962**	**0.841**	**-**
39–30-1	0.176	0.900	0.088	80.0	0.918	0.923	-
39–35-1	0.157	0.178	0.201	76.5	0.763	0.911	-
39-5-5–1	0.355	0.320	0.250	53.3	0.701	0.979	0.923
39–5-10–1	0.294	0.325	0.153	58.3	0.755	0.850	0.802
39–10-5–1	0.120	0.324	0.058	75.3	0.803	0.957	0.912
39–15-5–1	0.210	0.198	0.101	80.9	0.921	0.884	0.808
39–20-5–1	0.185	0.301	0.320	40.2	0.748	0.929	0.904
39–30-5–1	0.381	0.489	0.387	18.6	0.811	0.901	0.863
39–10-10–1	0.668	0.231	0.430	50.8	0.837	0.853	0.819
39–15-10–1	0.375	0.283	0.749	10.2	0.719	0.862	0.806
39–20-10–1	0.341	0.411	0.302	26.1	0.833	0.898	0.883
39–30-10–1	0.398	0.618	0.705	12.3	0.794	0.931	0.854
Powell-Beale							
39–10-1	0.379	0.314	0.451	21.8	0.740	0.921	-
39–30-1	0.185	0.301	0.085	73.5	0.860	0.938	-
39–35-1	0.217	0.274	0.152	53.7	0.802	0.889	0.810
39-5-5–1	0.239	0.214	0.149	91.4	0.964	0.852	0.835
39–5-10–1	0.177	0.303	0.283	42.3	0.711	0.837	0.781
39–10-5–1	0.222	0.209	0.200	38.6	0.698	0.826	0.793
39–15-5–1	0.222	0.141	0.425	87.4	0.915	0.912	0.875
39–20-5–1	0.154	0.332	0.259	31.6	0.732	0.969	0.800
39–30-5–1	**0.202**	**0.021**	**0.158**	**92.8**	**0.977**	**0.823**	**0.767**
39–10-10–1	0.025	0.121	0.107	85.6	0.903	0.952	0.931
39–15-10–1	0.285	0.145	0.839	84.2	0.881	0.851	0.849
39–20-10–1	0.392	0.371	0.264	64.2	0.902	0.882	0.803

accuracy obtained was 98.5 % and 97.1 % for ANNs 2 and 3, respectively. The efficiency of these structures is also confirmed by the AUC value obtained, which was the highest among all the tested structures.

3.3. Neurons saturation

The neuron's saturation was monitored for all the ANN structures tested in this work. This was done by calculating ψ_B for each hidden layer using Eqs. (4)–(6). All the results presented at Tables 5 and 6 are the ones with the lowest value of ψ_B. In Fig. 2 is shown the frequency distribution of the hidden layers outputs, and the ψ_B value, for the best structures of each ANN model. One can see that ANN 1 was the model with the higher saturation, following by ANN 2, and ANN 3, which presented the lowest values. Also, for the three models, the first layer presented higher ψ_B values.

3.4. 5-Fold Cross validation

A 5-fold cross validation was performed using the best structures obtained in each proposed model. In each case, all available data were divided in two sets: training (80 %) and validation (20 %). At Table 7 are shown the results obtained for the ANN 1, 2, and 3.

One can see that for all 5-fold run, the F_{OBJ} values were as low as the one presented at Tables 5 and 6. This result is corroborated by the average validation accuracy, which was similar to the obtained ANN model. Also, it suggests that using 70 % of all data to train the ANNs (as done previously) is sufficient to obtain a reliable model.

4. Discussion

In this work, several ANN structures were trained using different optimization algorithms to predict RVTE. Three models were proposed and the results showed that all of them perform this task with high accuracy.

For ANN 1, each algorithm presents different performances. In average, the Levenberg-Marquardt method presented the highest F_{OBJ} values and Resilient-Backpropagation/Powell-Beale presented the lowest. This behavior was expected and may be explained by the differences in the weight/bias actualization performed by each algorithm, which has a direct impact in the search for the function global minimum.

For ANN 2 and 3, the F_{OBJ} values followed no general tendency, as the number of neurons increases, and the best structure were 18 − 10-5 − 1 and 15 − 15-10 − 1, respectively. ANNs 2 and 3 were capable to predict the RVTE with a smaller network in comparison with ANN 1. We understand that this behavior is due the reduction of variables accomplished using PCA, which simplified the relation among the dependent and independent variables.

With the best results for ANN 1 (39 − 10-10 − 1), the confusion matrix for training, validation and test stages was constructed and it is shown at Table 8. In this table is also shown the confusion matrix for ANN 2 (18 − 10-5 − 1) and ANN 3 (15 − 15-10 − 1). It is possible to

Table 6
Results for ANNs 2 and 3 trained with Powell-Beale algorithm.

Structure	F_{OBJ}Training	F_{OBJ}Validation	F_{OBJ}Test	Validation + Test Accuracy (%)	Validation + Test AUC	$\psi_{B,1}$	$\psi_{B,2}$
ANN 2							
18–10-1	0.375	0.459	0.455	57.1	0.830	0.963	-
18–30-1	0.280	0.493	0.315	70.0	0.762	0.829	-
18–35-1	0.319	0.609	0.153	80.0	0.788	0.901	-
18–5-5–1	0.367	0.338	0.751	28.5	0.863	0.854	0.801
18–5-10–1	0.476	0.669	0.675	45.7	0.819	0.910	0.782
18–10-5–1	**0.236**	**0.115**	**0.001**	**98.5**	**0.983**	**0.678**	**0.625**
18–15-5–1	0.564	0.702	0.486	25.3	0.779	0.932	0.820
18–20-5–1	0.532	0.700	0.301	38.1	0.753	0.956	0.933
18–30-5–1	0.541	0.287	0.305	20.4	0.731	0.885	0.781
18–10-10–1	0.467	0.922	0.484	13.9	0.699	0.852	0.796
18–15-10–1	0.447	0.601	0.473	22.5	0.764	0.793	0.739
18–20-10–1	0.367	0.489	0.173	74.3	0.838	0.806	0.782
18–30-10–1	0.427	0.248	0.189	82.6	0.859	0.930	0.883
ANN 3							
15–10-1	0.599	0.412	0.571	10.0	0.830	0.955	-
15–30-1	0.629	0.401	0.457	4.3	0.748	0.902	-
15–35-1	0.345	0.509	0.353	76.0	0.785	0.853	-
15–5-5–1	0.307	0.384	0.385	57.1	0.812	0.791	0.753
15–5-10–1	0.614	0.501	0.258	47.1	0.870	0.753	0.689
15–10-5–1	0.339	0.343	0.532	84.2	0.813	0.863	0.723
15–15-5–1	0.124	0.245	0.578	82.9	0.812	0.802	0.723
15–20-5–1	0.624	0.560	0.329	50.0	0.665	0.763	0.703
15–30-5–1	0.433	0.576	0.411	52.8	0.865	0.932	0.893
15–10-10–1	0.416	0.560	0.132	75.7	0.858	0.891	0.830
15–15-10–1	**0.113**	**0.114**	**0.114**	**97.1**	**0.917**	**0.732**	**0.586**
15–20-10–1	0.232	0.753	0.542	80.0	0.825	0.893	0.821
15–30-10–1	0.428	0.287	0.833	57.1	0.652	0.854	0.783

observe that the prediction ANN 39 − 10-10 − 1 is very accurate. However, one can see that this model was the one with the highest number of false positives and false negatives. The number of false predictions for ANNs 2 and 3 was lower than for ANN 1 in all steps.

Although the good results obtained, it is important to keep track of the ANNs saturation. One can see in Fig. 2 that the best structure for all the models presented some degree of saturation. ANN 1 was the model with the highest saturation and ANN 3 presented de lowest. Since ψ_B for ANN 1 is close to 1.0, the generalization capacity of this ANN can be limited.

Saturation is an important measure to be monitored when training ANNs with binary classification. A saturated ANN could mean that the model is not capable to generalize its results and predict new cases. In this work, the results for the best ANNs showed that they presented some degree of saturation, especially in the first hidden layer. This behavior was also reported by Glorot and Bengio [37], but the authors did not stated any reason for this. In 2017, Kumar [39] presented the probability density function of the hyperbolic tangent function (*tanh* (x)), and it is possible to see that as the variance of x decreases, its result goes from bimodal distributed to normally distributed. The author also stated that the variance of the units inputs decreases as the information passes through the input layer, to the first hidden layer, and so on. Thus, it is expected that the first hidden layer presents a higher value of ψ_B.

Since the best structures obtained in this work have two hidden layers, Eq. (7) was used to calculate the importance of each input for all the ANN models. The detailed results are shown in Table S2, at the Supplemental File. Due the high number of inputs, the importance of each one in ANN 1 is distributed among all of them. In this model, the most important variable is the HDL (7.73 %), followed by RBC (7.64 %), RDW (5.55 %), D-dimer (5.48 %) and WBC (5.10 %). Is important to mention that when comparing the eighteen most important variables given by CWS, and the inputs of ANN 2 (given by Principal component Analysis [29]), one can see that 13 factors were indicated by both techniques.

For ANNs 2 and 3, since there are less inputs, the influence of the most important variable corresponds to 19.11 % and 14.91 %, respectively (WBC for ANN 2 and D-dimer for ANN 3). In these models, the first four factors have more influence than the others, accounting to 55 %. The different importance distribution is expected since ANN 2 and ANN 3 are relating two different set of inputs with the desired response.

Another important observation was that a limited number of patients (a total of 31) with pregnancy and postpartum, surgery, air travel longer than 6 h, cancer, and Antiphospholipid Syndrome were seen in the collected data. Although the cross-validation results showed that out model has good prediction capacity, this could be a limitation. Thus, we retrained the structure 15 − 15-10 − 1, obtained in ANN 3 model and excluded these patients from the dataset. This model considered 204 patients with a previous unprovoked VTE and patients with a first VTE provoked by hormone therapy. Other structures were not tested because the inputs were not modified. The results obtained showed that this structure is capable to predict the cases with high efficiency and without a significant loss of accuracy ($F_{OBJ,training}$ = 0.204, $F_{OBJ,validation}$ = 0.175, $F_{OBJ,test}$ = 0.061, accuracy = 93.7 %, AUC = 0.968).

Several studies that evaluate RVTE predictors can be found in the literature. However, developing clinical decision rules is a difficult task as usually only one factor at a time is considered and the number of patients is limited. To overcome these drawbacks, some methods to calculate the risk of RVTE were proposed. Dash [2], Vienna [3, 4] and HERDOO2 [5] scores are the three methods available to calculate the risk for RVTE after three months anticoagulation therapy. Mohamed [40] proposed several models obtained via regularized regressions using data from the MEGA study.

Dash score calculates RVTE risk based on D-dimer level, age, sex, and use of hormonal therapy. Vienna score calculate the risk based on sex, location of the first VTE (distal/proximal VTE or pulmonary embolism), and D-dimer level. The score developed by Rodger et al. [5] is only applied in women and takes age, D-dimer level, body mass index, and signs of hyperpigmentation, oedema or redness of either leg into consideration and states that women with none or one of the criteria can safely discontinue anticoagulants. All scores can be used only if the patient had a first unprovoked VTE.

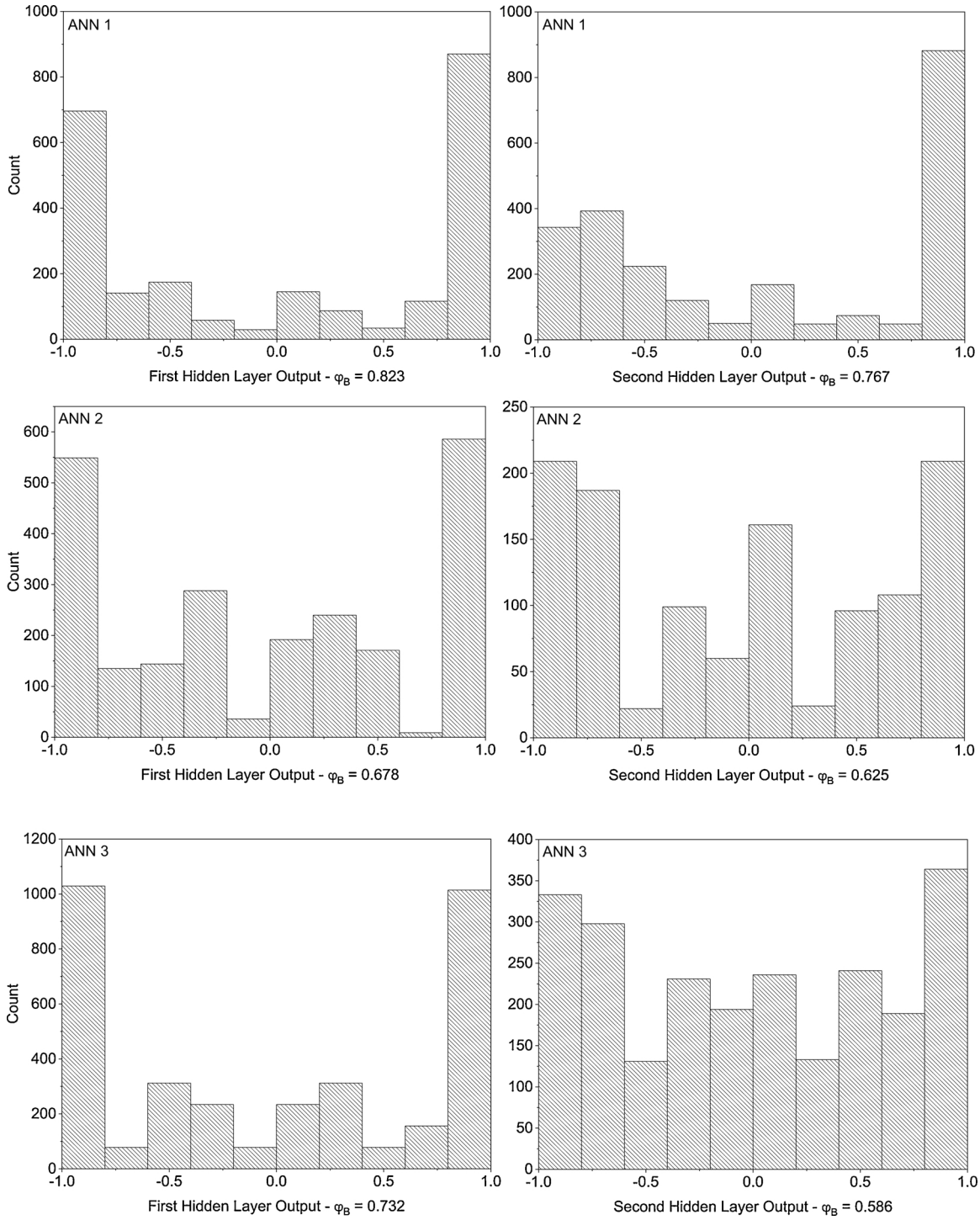

Fig. 2. Hidden unit output values frequency distributions.

Reviews concerning the applicability of these scores can be found in literature [6, 7]. In a recent study, Ensor et al. [7] extensively reviewed the advantages and disadvantages of each score. According to the authors, all three models were proposed using similar population data and the Dash model may have some biased conclusions, since patients with hormone intake at the time of event were also included. Furthermore, all scores suffer from developing limitations and lack of external validation.

External validation of any given mathematical model is crucial for practical uses. Marcucci et al. [41] performed a validation study of the Vienna score [3] and reported that this score is capable of distinguishing a patient's risk. However, they found that the model tends to underestimate the risk of RVTE at 12 months after ending coagulation therapy. Recently, Tritschler et al. [42] aimed to validate the updated Vienna model [4] for elderly patients. Their results showed that it was not possible to identify patients with low-risk of RVTE and suggested that a specific validation should be performed. Rodger et al. [43] performed a validation of their proposed HERDOO2 rule and showed that indeed it works for these specific cases.

Some patients develop VTE due the presence of transient risk factors, which can be reversible, such as: surgery, trauma, pregnancy, etc, or prolonged, such as: cancer. It is widely accepted that these patients

Table 7
Results obtained in ANNs 5-fold cross validation.

k-Fold Run Index	F_{OBJ} Training	F_{OBJ} Validation	Validation Accuracy (%)	Validation AUC
ANN 1 – 39 – 10-10 – 1				
1	0.192	0.516	90.9	0.959
2	0.232	0.196	95.4	0.912
3	0.188	0.356	93.2	0.998
4	0.221	0.513	95.4	0.999
5	0.201	0.386	90.9	0.938
Average			93.2	0.961
ANN 2 – 18 – 10-5 – 1				
1	0.217	0.151	98.7	0.958
2	0.197	0.246	95.7	0.914
3	0.173	0.365	94.2	0.912
4	0.161	0.464	97.8	0.920
5	0.224	0.123	98.4	0.969
Average			96.9	0.934
ANN 3 – 15 – 15-10 – 1				
1	0.154	0.157	95.8	0.984
2	0.114	0.377	93.5	0.953
3	0.147	0.044	95.8	1.000
4	0.130	0.030	97.9	1.000
5	0.117	0.159	97.9	0.999
Average			96.2	0.987

Table 8
Confusion matrix obtained using the best structure of ANNs 1, 2, and 3.

	Training		Validation		Test	
ANN 1 – 39 – 10-10 – 1						
	RVTE	Non-RVTE	RVTE	Non-RVTE	RVTE	Non-RVTE
RVTE	32	7	6	0	6	3
Non-RVTE	4	122	1	28	1	25
ANN 2 – 18 – 10-5 – 1						
RVTE	28	3	10	0	7	0
Non-RVTE	3	131	0	25	1	27
ANN 3 – 15 – 15-10 – 1						
RVTE	30	2	6	0	11	1
Non-RVTE	1	132	1	28	0	23

have a very low risk of RVTE in comparison with those with unprovoked VTE [44]. However, RVTE in these conditions is not yet well understood.

In a review study, Iorio et al. [44] stated that the risk of RVTE appears to be low in patients with VTE provoked by a reversible risk factor and by surgery, and intermediate in patients with VTE provoked by a nonsurgical risk factor. Furthermore, they affirm that both risks are very low when compared to the patients with a previous unprovoked VTE. Cosmi et al. [45] followed 296 patients with a previous provoked VTE in a 2-year study and during their study, 15 patients presented RVTE (a rate of 5.1 %). Of those 15, 10 presented higher D-dimer level after stopping anticoagulation. For these patients, in 12 months, the authors reported a 7–11 % rate of recurrence, which is similar to the annual risk of RVTE for patients with unprovoked VTE. Therefore, this subject should be addressed more carefully.

This work aimed to develop a new model to predict RVTE in patients with both provoked and unprovoked previous VTE using different clinical, inherited, acquired and molecular risk factors. ANN 1 included patient differentiation regarding sex, age, as well as previous provoked or unprovoked VTE. In ANN 2, only variables pointed out as important by PCA were considered. ANN 3 considered inputs that are simple and fast to obtain. All models were successful for RVTE prediction, since the results showed that accuracy were very close to 100 %. Considering the saturation measure calculated, we would indicate that ANNs 2 and 3 are promising models for future practical application.

The models developed in this work also require external validation to offer generalization information. Until this moment, this work showed that ANNs are promising and the models obtained should be improved by increasing the number of patients. The ANNs saturation degree can also be improved by presenting a higher number of examples, which is important to guarantee its generalization capacity. Once these ANNs are validated, they could be used for clinicians in day life practice. Validated models can also be used to evaluate the importance, and the sign effect, of each input parameter using Design of Experiment (DoE) techniques. Using the ANN 1, with the input combinations from the DoE planning, one can determine which factors are statistically significant to predict RVTE with a *t*-Student test. Then, using these new selected factors, one can obtain another ANN model with other inputs. The result also could be used to improve the number of inputs along with the PCA results. Finally, since the equation of an ANN is a series of multiplication and sum of matrices, validated ANNs could be used in practical medicine by programming then in an electronic spreadsheet, or using an smartphone application with an easy user interface.

In summary, this work showed that the use of PCA to select ANN inputs before training is an effective approach and can be used to develop predictive models for other diseases. Also, that ANN models is an efficient equation for RVTE risk prediction. These equations can be developed including numerous factors and have the power to extract the interactions among them with no loss of accuracy. Future work will focus on the external validation of the models with a different database containing a higher number of patients, and the use of neuro-fuzzy networks, supporting vector machines, echo-state networks, and regularization regression techniques to predict RVTE.

5. Conclusions

In this work, three ANN models were developed to predict RVTE in patients with provoked and unprovoked first VTE. Two strategies were employed: the first included 39 input factors and the second, 18 inputs obtained using PCA. A third model was proposed combining the inputs of model 2 with practical aspects. Different ANN structures and optimization algorithms were tested in order to search for the best models. The results showed that the ANNs are efficient to predict RVTE for all sets of input factors. The best models presented high accuracy and the 5-fold cross validation confirmed their prediction capacity. Finally, this work showed that ANN is a powerful tool that can model a complex phenomenon such as RVTE and with proper external validation could be used by clinicians in daylife basis.

Authors' statement

Each author contributed to: 1) development, review and commented on each draft 2) study protocol 3) data acquisition 4) interpretation and discussion. All authors agreed to the publication of the article presented in this form.

Ethical approval

The Ethical approval to report this case was obtained from Comite de Ética da Universidade Estadual de Campinas (Approval number: 88970218.0.0000.5404).

Summary

What was already known on the topic:

- VTE recurrence can be predicted using Scores;
- Application of VTE recurrence scores are limited;
- Neural networks are already used to predict diseases;
- There is no neural network model to predict VTE recurrence.

What this study added to our knowledge

- Neural networks can be used to predict VTE recurrence accurately;
- Neural networks can learn about VTE recurrence using different set

of inputs;

- Neural networks can be more accurate than scores models.

;1;

Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships which have, or could be perceived to have, influenced the work reported in this article.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijmedinf.2020.104221.

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